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解: $t < 0$ 时电路为零状态, 由换路定律得:

$$u_C(0_+) = u_C(0_-) = 0$$

$t > 0$ 时为简化计算, 先将 ab 左边电路化为戴维南电路形式。

当 ab 端开路时, 由 $i + 2i = 0$, 得 $i = 0$ 所以开路电压

$$u_{OC} = u_s = 10\sqrt{2}\cos(100t)\text{V}$$

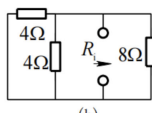
当 ab 端短路时,

$$i_{SC} = i + 2i = 3i = 3 \times \frac{u_s}{3\Omega}$$

故等效电阻

$$R_i = \frac{u_{OC}}{i_{SC}} = 1\Omega,$$

$t > 0$ 时等效电路如图(b)所示。



电路时间常数为

$$\tau = R_i C = 0.01\text{s}.$$

用相量法计算强制分量 u_{CP} :

$$\dot{U}_{CP} = \frac{1/(j\omega C)}{1 + 1/(j\omega C)} \times \dot{U}_{OC} = \frac{-j}{1-j} \times 10\angle 0^\circ = 5\sqrt{2}\angle -45^\circ\text{V}$$

$$u_{CP}(t) = 10\cos(100t - 45^\circ)\text{V}$$

$$u_{CP}(0_+) = 10\cos(-45^\circ) = 5\sqrt{2}\text{V}$$

由三要素公式得:

$$u_C(t) = u_{CP}(t) + [u_C(0_+) - u_{CP}(0_+)]e^{-t/\tau} = [10\cos(100t - 45^\circ) - 5\sqrt{2}e^{-100t}]\text{V}$$

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解: 由于 i_s 为指数函数, 故须列写关于 i 的微分方程来计算 i 的强制分量。

由换路定律得:

$$i_L(0_+) = i_L(0_-) = 3A$$

$$i(0_+) = i_s(0_+) - i_L(0_+) = 5 - 3 = 2A \quad (1)$$

根据 KVL

$$L \frac{di_L}{dt} - 3i - 2i = 0$$

将 $i_L = i_s - i$ 代入上式化简得

$$L \frac{di}{dt} + 5i = L \frac{di_s}{dt} = -25e^{-10t}$$

$$\frac{di}{dt} + 10i = -50e^{-10t} \quad (2)$$

由式(1)中得时间常数 $\tau = 1/10 = 0.1s$ 等于电流源衰减系数的倒数, 故设强制

分量为

$$i_p(t) = A_1 t e^{-10t}, \text{ 代入式(2)解得 } A_1 = -50。$$

设齐次分量为 $i_h(t) = A_2 e^{-10t}$, 则电流 i 的完全解答为:

$$i(t) = i_p(t) + i_h(t) = -50t e^{-10t} + A_2 e^{-10t} \quad (3)$$

由初始条件确定待求系数 A_2 。由式(3)及式(1)得 $i(0_+) = A_2 = 2$, 即 $A_2 = 2$ 。

因此

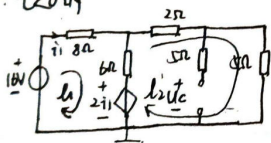
$$i(t) = [2e^{-10t} - 50te^{-10t}] A$$

强制分量为 $-50te^{-10t}$, 自由分量为 $2e^{-10t}$ 。

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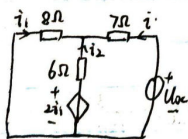
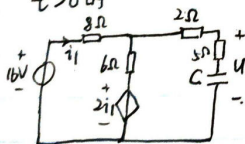
8.19 解: $t < 0$ 时



$$\begin{cases} (8+6\Omega)i_1 - 6i_2 = 16V - 2i_1 \\ (6\Omega + 2\Omega + 5\Omega)i_2 - 6i_1 = 2i_1 \\ u_C(0^+) = u_C(0^-) = 5\Omega \cdot i_2 \end{cases}$$

解得 $u_C(0^+) = u_C(0^-) = 4V$.

$t > 0$ 时



$$8i_1 = 6i_2 - 2i_1,$$

$$R_{eq} = 7\Omega + \frac{8i_1}{i_1 + i_2} = 10\Omega, Z = R_{eq} \cdot C = 0.1s$$

$$t \rightarrow \infty \text{ 时 } 16V - 2i_1 = (8\Omega + 6\Omega)i_1$$

$$\text{故 } i_1 = 1A, u_C(\infty) = 16V - 8i_1 = 8V.$$

$$\text{由三要素公式, } u_C(t) = 8 + [4 - 8]e^{-\frac{t}{0.1}} = 8 - 4e^{-10t} V.$$

$$\text{由 } u(\infty) = u_C(\infty) = 8V, u(0^+) = u_C(0^+) + 5\Omega \cdot 0.4A = 6V.$$

$$u(t) = u(\infty) + [u(0^+) - u(\infty)]e^{-\frac{t}{\tau}} = 8 + (6 - 8)e^{-10t} = 8 - 2e^{-10t} V.$$

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解: 当 $t < 0$ 时, 电容开路, 电感短路, 列写节点电压方程如下

$$\left(\frac{1}{3} + \frac{1}{2} + \frac{1}{6} + \frac{1}{3}\right)u_C(0_-) - \frac{4}{3} - \left(\frac{1}{3} + \frac{1}{6}\right) \times 16 = 0$$

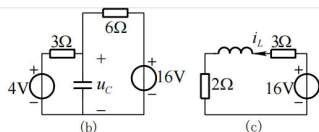
解得

$$u_C(0_-) = 7V$$

由换路定律得:

$$u_C(0_+) = u_C(0_-) = 7V, \quad i_L(0_+) = i_L(0_-) = 3A$$

换路后构成两个一阶电路, 如图 (b) 和(c)所示。



在图(b) 电路中, 稳态时电容开路, 所以

$$u_C(\infty) = \frac{16-4}{3+6} \times 3 + 4 = 8V$$

等效电阻

$$R_i = \frac{3 \times 6}{3+6} = 2\Omega$$

时间常数

$$\tau_C = 2 \times 1 = 2s$$

由三要素公式得

$$u_C(t) = (8 - e^{-0.5t})V$$

在图(c)电路中, 稳态时电感短路, 所以

$$i_L(\infty) = \frac{16}{2+3} = 3.2A$$

时间常数

$$\tau_L = \frac{2}{2+3} = 0.4s,$$

由三要素公式得:

$$i_L(t) = (3.2 - 0.2e^{-2.5t})A$$

开关电压

$$u(t) = u_C - 2\Omega \times i_L = (1.6 - e^{-0.5t} + 0.4e^{-2.5t})V \quad (t > 0)$$

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8.23解: 1) 由全响应 $u_C(t) = 2 + 3e^{-2t} + 5e^{-t}$ V 得

$$U_{S1} = 2V, U_{S2} = 3V, u_C' = u_C(0^+)e^{-t}, \tau = 1s$$

$$\text{令 } t=0, u_C(0^+) = 10V, \text{ 故 } u_C' = 10e^{-t} V.$$

$$(2) U_{S1} \text{ 单独作用: } u_{Cp} = U_{S1} = 2V, u_C' = 2V + (0 - 2V)e^{-t} = 2(1 - e^{-t})V.$$

$$U_{S2} \text{ 单独作用: } u_{Cp} = U_{S2} = 3e^{-2t}V, u_{Cp}(0^+) = 3V.$$

$$u_C'' = 3e^{-2t} + (0 - 3V)e^{-t} = 3(e^{-2t} - e^{-t})V.$$

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8.24

$$i(0_+) = i(0_-) = 0$$

稳态值

$$i(\infty) = \frac{32}{12+8} = 1.6 \text{ A}$$

串联等效电感

$$L = L_1 + L_2 - 2M = 0.2 + 0.4 - 2 \times 0.1 = 0.4 \text{ H}$$

等效电阻

$$R = 12 + 8 = 20 \Omega$$

时间常数

$$\tau = \frac{L}{R} = \frac{0.4}{20} = \frac{1}{50} \text{ s}$$

由三要素公式得:

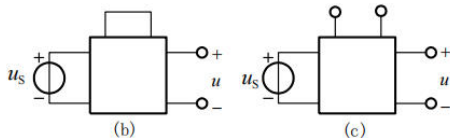
$$i(t) = 1.6(1 - e^{-50t}) \text{ A} \quad t \geq 0$$

$$u(t) = (L_2 - M) \frac{di}{dt} = 0.3 \times 1.6 \times 50 e^{-50t} = 24 e^{-50t} \text{ V} \quad (t > 0)$$

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解：由题接电容时的零状态响应，可得 $t=0_+$ 和 $t \rightarrow \infty$ 时的计算电路，分别如图(b)和(c)所示。



由于电感对直流稳态相当于短路，零状态电感在换路瞬间相当于开路，故接电感在 $t=0_+$ 和 $t \rightarrow \infty$ 时的计算电路分别与接电容时 $t \rightarrow \infty$ 和 $t=0_+$ 时的情况相同。所以接 L 时，初始值 $u(0_+)=10\text{V}$ ，稳态值 $u(\infty)=5\text{V}$ 。

由接电容时的响应得时间常数

$$\tau_c = 0.5 = R_l C, \text{ 所以 } R_l = \frac{\tau_c}{C} = 50\Omega$$

接电感后， R_l 不变，故时间常数

$$\tau_L = \frac{L}{R_l} = 0.1\text{s}$$

将上述初始值、稳态值和时间常数代入三要素公式得

$$u(t) = [5 + 5e^{-10t}] \varepsilon(t) \text{V}$$

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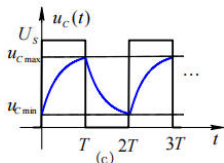
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解: 达到稳定后开始计时, 在 $0 \leq t \leq T$ 内, 电容从最小值 $u_{C\min}$ 开始充电, 在 $t = T$ 时刻达到最大值。初始值 $u_C(0_+) = u_{C\min}$, 特解 $u_{Cp}(t) = U_S$, $u_{Cp}(0_+) = U_S$, 时间常数 $\tau = RC$ 。

由二要素公式得:

$$u_C(t) = U_S + (u_{C\min} - U_S)e^{-t/\tau} \quad 0 \leq t \leq T \quad (1)$$

在 $T \leq t \leq 2T$ 内, 电容由最大值 $u_{C\max}$ 开始放电, 在 $t = 2T$ 时达到最小值。波形如图(c)所示。



此时间电路为零输入响应, 电容电压为:

$$u_C(t) = u_{C\max} e^{-(t-T)/\tau} \quad T \leq t \leq 2T \quad (2)$$

由式(1)得:

$$u_C(T) = U_S + (u_{C\min} - U_S)e^{-T/\tau} = u_{C\max} \quad (3)$$

由式(2)得:

$$u_C(2T) = u_{C\max} e^{-T/\tau} = u_{C\min} \quad (4)$$

通过联立求解式(3)和(4)便可证得

$$u_{C\max} = \frac{U_S}{1 + e^{-T/\tau}}, \quad u_{C\min} = \frac{U_S e^{-T/\tau}}{1 + e^{-T/\tau}}$$

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解: (1) 当 $i_s = \varepsilon(t)$ A 时, 先求 ab 两端的戴维南等效电路。ab 端开路时, 根据

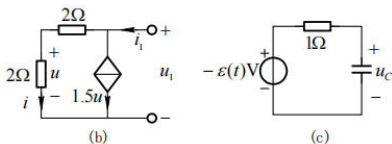
图(a)电路, 由 KCL 得:

$$\frac{u}{2} + 1.5u = i_s = \varepsilon(t) \Rightarrow u = 0.5\varepsilon(t)\text{V}$$

开路电压

$$u_{oc} = u - 2 \times 1.5u = -2 \times u = -1\varepsilon(t)\text{V}$$

求等效电阻的电路如图(b)所示。



$$u_i = (2+2)i = (2+2) \times \frac{u}{2} = 2u$$

$$i_i = i + 1.5u = \frac{1}{2}u + 1.5u = 2u$$

等效电阻

$$R_i = \frac{u_i}{i_i} = 1\Omega$$

戴维南等效电路如图 (c) 所示。时间常数

$$\tau = R_i C = 0.1\text{s}。$$

根据三要素公式得 u_c 的单位阶跃特性为:

$$s(t) = -1(1 - e^{-10t})\varepsilon(t) \text{ V}$$

(2) 单位冲激特性为:

$$h(t) = \frac{ds(t)}{dt} = -10e^{-10t}\varepsilon(t) \text{ V/s}$$

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解：电压源为单位冲激函数，不能直接求其响应，而应先求单位阶跃响应，再对其求导得到单位冲激响应。为此先求 ab 端左侧的戴维南等效电路。当 ab 端开路时，

$$i = 3i \Rightarrow i = 0$$

开路电压

$$u_{oc} = u_s$$

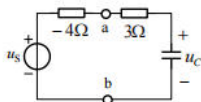
当 ab 端短路时，短路电流

$$i_{sc} = i - 3i = -2i = -2 \times \frac{u_s}{8\Omega}$$

等效电阻

$$R_i = \frac{u_{oc}}{i_{sc}} = -4\Omega$$

图(a)的等效电路如图(b)所示。时间常数



(b)

$$\tau = (3 - 4) \times 0.02 = -0.02 \text{ s}$$

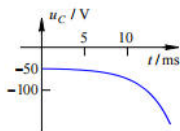
由三要素公式得 u_c 的单位阶跃特性为：

$$s(t) = (1 - e^{50t})\varepsilon(t)$$

u_c 的单位冲激响应为：

$$u_c(t) = 1\text{Wb} \times h(t) = \frac{ds(t)}{dt} = -50e^{50t}\varepsilon(t) \text{ V}$$

其波形如图 (c) 所示。



(c)

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解:

$$u_1(0_-) = 6V, \quad u_2(0_-) = 10V$$

$t=0$ 时开关接通, 两电压原始值不等的电容相并联, 电容电压将发生跃变。利

用两正极板电荷之和在开关动作前后瞬间相等来计算 $u_2(0_+)$:

$$\begin{cases} 0.2u_1(0_+) + 0.3u_2(0_+) = 0.2u_1(0_-) + 0.3u_2(0_-) \\ u_1(0_+) = u_2(0_+) \end{cases}$$

解得

$$u_1(0_+) = u_2(0_+) = 8.4V$$

稳态值

$$u_2(\infty) = \frac{6}{6+6} \times 12V = 6V$$

时间常数

$$\tau = RC = (6/2) \times (0.2+0.3) = 1.5s$$

由三要素公式得:

$$u_2(t) = u_2(\infty) + [u_2(0_+) - u_2(\infty)]e^{-t/\tau} = (6 + 2.4e^{-t/1.5})V \quad (t > 0)$$

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8.28

8.28 解: 接通时电容电压无跃变, $u_1(0+) = u_1(0-) = 20\text{V}$, $u_2(0+) = u_2(0-) = 0$.

根据电荷守恒,
$$\begin{cases} 3F u_1(\infty) + 6F u_2(\infty) = 3F \cdot u_1(0+) + 6F u_2(0+) \\ u_1(\infty) = u_2(\infty) \end{cases}$$

解得 $u_1(\infty) = u_2(\infty) = \frac{20}{3}\text{V}$, $\tau = RC = 10\Omega \cdot \frac{3F \cdot 6F}{3F + 6F} = 20\text{s}$.

故 $u_1(t) = u_1(\infty) + [u_1(0+) - u_1(\infty)]e^{-t/\tau} = \frac{20}{3} + \frac{40}{3}e^{-0.05t}\text{V}$.

$u_2(t) = u_2(\infty) + [u_2(0+) - u_2(\infty)]e^{-t/\tau} = \frac{20}{3} - \frac{20}{3}e^{-0.05t}\text{V}$.

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8.39

解:

$$i(0_+) = i(0_-) = \frac{U_s}{R_2}, \quad u_C(0_+) = u_C(0_-) = 0$$

根据 KVL, 列写方程如下:

$$R_1 C \frac{du_C}{dt} + u_C = u = U_s \quad (1)$$

$$L \frac{di}{dt} + R_2 i = U_s - u = 0 \quad (2)$$

由式(1)解得

$$u_C(t) = U_s(1 - e^{-t/R_1 C}) \quad (3)$$

$$i = C \frac{du_C}{dt} = \frac{U_s}{R_1} e^{-t/R_1 C} \quad (4)$$

由式(2)又解得

$$i(t) = \frac{U_s}{R_2} e^{-\frac{R_2}{L} t} \quad (5)$$

由式(4)和式(5)相等解得

$$R_1 = R_2 = \sqrt{\frac{L}{C}}$$

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解: 由 KVL 得:

$$u_L + u_C = L \frac{di_L}{dt} + u_C = 20V \quad (1)$$

由 KCL 得:

$$-i_L + i_C + i_C = -i_L + \frac{8}{7}u_C + \frac{1}{7} \frac{du_C}{dt} = 0 \quad (2)$$

方程(2)对 t 求导, 再将方程(1)代入, 经整理得:

$$\frac{d^2 u_C}{dt^2} + 8 \frac{du_C}{dt} + 7u_C = 140 \quad (3)$$

因为

$$i_L(0_+) = i_L(0_-) = \frac{20V}{7/8\Omega} = \frac{160}{7}A$$

所以 u_C 及其导数的初始条件为

$$\begin{cases} u_C(0_+) = u_C(0_-) = 10V \\ \left. \frac{du_C}{dt} \right|_{t=0_+} = \frac{1}{C} [i_L(0_+) - \frac{u_C(0_+)}{7/8}] = 80 \end{cases} \quad (4)$$

微分方程(3)的特征方程为:

$$p^2 + 8p + 7 = 0$$

解得

$$p_1 = -1, \quad p_2 = -7$$

稳态时, $u_C(\infty) = 20V$, 所以特解 $u_{Ca}(t) = 20$

设其完全解答为:

$$u_C(t) = 20 + A_1 e^{-t} + A_2 e^{-7t} \quad (5)$$

由初始条件(4)得

$$\begin{cases} u_C(0_+) = A_1 + A_2 + 20 = 10 \\ \left. \frac{du_C}{dt} \right|_{t=0_+} = -A_1 - 7A_2 = 80 \end{cases}$$

解得

$$A_1 = 1.67, \quad A_2 = -11.67$$

所以将 A_1 、 A_2 代入(5)得:

$$u_C(t) = [20 + 1.67e^{-t} - 11.67e^{-7t}]V$$

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9.1

解:

(1)

$$F(s) = \int_0^{\infty} t \varepsilon(t) e^{-st} dt = -\frac{t}{s} e^{-st} \Big|_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt = -\frac{1}{s^2} e^{-st} \Big|_0^{\infty} = \frac{1}{s^2}$$

(2)

$$\begin{aligned} F(s) &= \int_0^{\infty} t e^{-\alpha t} \varepsilon(t) e^{-st} dt = -\frac{t}{s+\alpha} e^{-(s+\alpha)t} \Big|_0^{\infty} + \frac{1}{s+\alpha} \int_0^{\infty} e^{-(s+\alpha)t} dt \\ &= -\frac{1}{(s+\alpha)^2} e^{-(s+\alpha)t} \Big|_0^{\infty} = \frac{1}{(s+\alpha)^2} \end{aligned}$$

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9.2

解: (a)

$$F(s) = \frac{2s+1}{s^2+5s+6} = \frac{A_1}{s+2} + \frac{A_2}{s+3}$$

$$A_1 = \frac{2s+1}{s+3} \Big|_{s=-2} = -3, \quad A_2 = \frac{2s+1}{s+3} \Big|_{s=-3} = -3$$

所以

$$f(t) = \mathcal{L}^{-1} \left\{ \frac{-3}{s+2} + \frac{5}{s+3} \right\} = -3e^{-2t} + 5e^{-3t}$$

(b)

$$F(s) = \frac{s^3+5s^2+9s+7}{(s+1)(s+2)} = s+2 + \frac{s+3}{(s+1)(s+2)} = s+2 + \frac{A_1}{s+1} + \frac{A_2}{s+2}$$

$$A_1 = \frac{s+3}{s+2} \Big|_{s=-1} = 2, \quad A_2 = \frac{s+3}{s+1} \Big|_{s=-2} = -1$$

所以

$$f(t) = \mathcal{L}^{-1} \left\{ s+2 + \frac{2}{s+1} + \frac{-1}{s+2} \right\} = \delta'(t) + 2\delta(t) + 2e^{-t} - e^{-2t}$$

(c)

$$F(s) = \frac{3}{s^2+2s+6} = \frac{(3/\sqrt{5}) \times \sqrt{5}}{(s+1)^2 + (\sqrt{5})^2}$$

查表得

$$f(t) = \frac{3}{\sqrt{5}} e^{-t} \sin(\sqrt{5}t)$$

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9.4

解:

$$U_s(s) = \mathbf{L}[e^{-2t}\varepsilon(t)] = \frac{1}{s+2}$$

由运算电路 (略) 并利用分压公式求得电容电压象函数为:

$$U(s) = \frac{\frac{2 \times 2s}{2+2s}}{1 + \frac{2 \times 2s}{2+2s}} \times U_s(s) = \frac{(2/3)s}{(s+1/3)(s+2)} = \frac{A_1}{s+1/3} + \frac{A_2}{s+2}$$

式中

$$A_1 = \frac{(2/3)s}{s+2} \Big|_{s=-1/3} = -\frac{2}{15} \text{Vs}, \quad A_2 = \frac{(2/3)s}{s+1/3} \Big|_{s=-2} = 0.8 \text{Vs}$$

所以

$$u(t) = [0.8e^{-2t} - (2/15)e^{-t/3}] \text{V}$$

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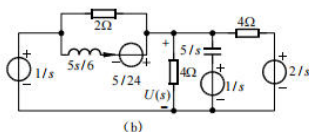
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9.6

解: 由图(a)得:

$$u(0_-) = U_1 = 1\text{V}, \quad i_L(0_-) = U_1 / R_2 = 0.25\text{A}$$

运算电路如图 11.9(b)所示。



列写节点电压方程如下:

$$\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{4} + 0.2s + \frac{1}{5s/6}\right)U(s) - \left(\frac{1}{2} + \frac{1}{5s/6}\right) \times \frac{1}{s} = \frac{2/s}{4} + \frac{1/s}{5/s} + \frac{5/24}{5s/6}$$

解得:

$$U(s) = \frac{s^2 + 6.25s + 6}{s(s^2 + 5s + 6)} = \frac{A_1}{s} + \frac{A_2}{s+2} + \frac{A_3}{s+3}$$

各待定系数为

$$A_1 = \frac{s^2 + 6.25s + 6}{(s+2)(s+3)} \Big|_{s=0} = 1\text{Vs}$$

$$A_2 = \frac{s^2 + 6.25s + 6}{s(s+3)} \Big|_{s=-2} = 1.25\text{Vs}$$

$$A_3 = \frac{s^2 + 6.25s + 6}{s(s+2)} \Big|_{s=-3} = 1\text{Vs}$$

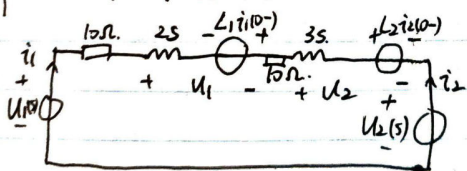
所以

$$u(t) = \mathcal{L}^{-1}\{U(s)\} = (1 + 1.25e^{-2t} - 1.25e^{-3t})\text{V}$$

日期:

9.13

9.13 解: $t < 0$ 时 $i_{L1}(0+) = i_{L1}(0-) = 2A$, $i_{L2}(0+) = i_{L2}(0-) = 1A$.



$$U_1(s) = \frac{20}{s} V_s, \quad U_2(s) = \frac{10}{s} V_s.$$

$$\text{令 } i_1 = -i_2 = i, \text{ 则}$$

$$(20\Omega + 5S)I(s) + L_2 i_{L2}(0-) + U_2(s) = U_1(s) + L_1 i_{L1}(0-).$$

$$\text{解得 } I(s) = \frac{10+s}{5s(s+4)} = \frac{1}{5} \left(\frac{\frac{5}{s}}{s} + \frac{-\frac{3}{s+4}}{s+4} \right), \quad \text{故 } i(t) = \left(\frac{1}{2} - \frac{3}{10} e^{-4t} \right) \varepsilon(t) A.$$

$$U_1(s) = I(s) \cdot 2S - L_1 i_{L1}(0-) = -\frac{18}{s} + \frac{12}{s} \cdot \frac{1}{s+4} = -3.6 + 2.4 \cdot \frac{1}{s+4} V.$$

$$u_1(t) = -3.6 \delta(t) + 2.4 e^{-4t} \varepsilon(t) V.$$

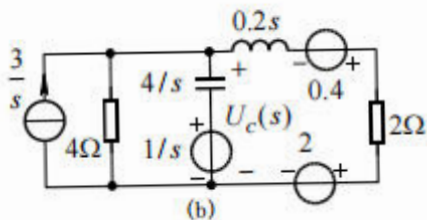
$$U_2(s) = I(s) \cdot 3S + L_2 i_{L2}(0-) = \frac{3}{s} \frac{s+10}{s+4} + 3 = 3.6 + 3.6 \frac{1}{s+4}$$

$$\text{故 } u_2(t) = 3.6 \delta(t) + 3.6 e^{-4t} \varepsilon(t) V.$$

日期: /

9.14

解：画出运算电路如图(b)所示，列写节点电压方程如下：



$$(0.25s + 0.25 + \frac{1}{2 + 0.2s})U_C(s) = \frac{3}{s} + \frac{1}{s} \times 0.25s + \frac{2 - 0.4}{2 + 0.2s}$$

解得：

$$U_C(s) = \frac{s^2 + 54s + 120}{s(s+5)(s+6)} = \frac{A_1}{s} + \frac{A_2}{s+5} + \frac{A_3}{s+6}$$

式中

$$A_1 = \frac{s^2 + 54s + 120}{(s+5)(s+6)} \Big|_{s=0} = 4Vs$$

$$A_2 = \frac{s^2 + 54s + 120}{s(s+6)} \Big|_{s=-5} = 25Vs,$$

$$A_3 = \frac{s^2 + 54s + 120}{s(s+5)} \Big|_{s=-6} = -28Vs$$

反变换得

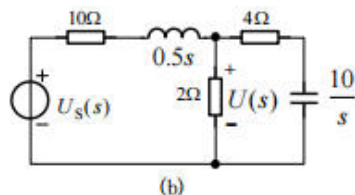
$$u_C(t) = [4 + 25e^{-5t} - 28e^{-6t}]V \quad t > 0$$

日期:

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9.28

解：运算电路如图(b)所示。



列写节点电压方程如下：

$$\left(\frac{1}{2} + \frac{1}{10 + 0.5s} + \frac{1}{4 + 10/s}\right)U(s) = \frac{U_s(s)}{10 + 0.5s}$$

解得

$$H(s) = \frac{U(s)}{U_s(s)} = \frac{8s + 20}{3s^2 + 73s + 120}$$

故

$$H(j\omega) = \frac{8 \times j\omega + 20}{3 \times (j\omega)^2 + 73 \times j\omega + 120} = \frac{20 + j8\omega}{120 - 3\omega^2 + j73\omega}$$

当

$$u_s = (100\sqrt{2} \cos 10t) \text{ V 时, } \dot{U}_s = 100 \text{ V, } \omega = 10 \text{ rad/s}$$

$$\dot{U} = H(j10) \times \dot{U}_s = \frac{20 + j80}{120 - 300 + j730} \times 100 \text{ V} = 10.967 \angle -27.89^\circ \text{ V}$$

正弦稳态电压

$$u = 10.967\sqrt{2} \cos(10t - 27.89^\circ) \text{ V}$$

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9.29

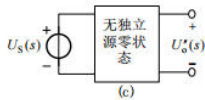
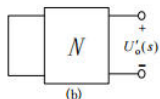
解: 对图示电路, 在复频域中, 根据叠加定理和齐性定理, 全响应的一般表

达式可以写成

$$U(s) = U'_o(s) + U''_o(s) = U'_o(s) + H(s)U_s(s) \quad (1)$$

其中 $U'_o(s)$ 是仅由二端口网络内部电源及初始储能作用时产生的响应分量,

如图(b)所示; $U''_o(s)$ 则是仅由 $U_s(s)$ 单独作用时产生的响应分量, 如图(c)所示。



根据网络函数定义得

$$U''_o(s) = H(s)U_s(s)。$$

对题给激励及响应进行拉普拉斯变换, 代入式(1)得

$$\begin{cases} \frac{8}{s} + \frac{2}{s+0.2} = U'_o(s) + H(s) \times \frac{6}{s} \\ \frac{11}{s} - \frac{1}{s+0.2} = U'_o(s) + H(s) \times \frac{12}{s} \end{cases}$$

解得

$$\begin{cases} H(s) = \frac{0.1}{s+0.2} \\ U'_o(s) = \frac{10s+1}{s(s+0.2)} \end{cases}$$

当 $u_s = 6e^{-5t}\varepsilon(t)$ V 即 $U_s(s) = \frac{6}{s+5}$ 时, 响应象函数

$$U_o(s) = U'_o(s) + H(s) \times \frac{6}{s+5} = \frac{5}{s} + \frac{5.125}{s+0.2} - \frac{0.125}{s+5}$$

反变换得

$$u_o(t) = \mathcal{L}^{-1}\{U_o(s)\} = (5 + 5.125e^{-0.2t} - 0.125e^{-5t}) \varepsilon(t) \text{ V}$$

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9.31

解: 电路的全响应等于强制分量与自由分量之和, 强制分量一般由外加激励决定, 自由分量的函数形式取決与网络函数极点性质。故本题全响应可以写成

$$u_o = u_{op} + u_{oh} = u_{op} + Ae^{-t} + Be^{-2t} \quad (1)$$

当激励为正弦量时, 响应的强制分量也为同频率的正弦量, 可用相量法求出。
频域形式的网络函数为

$$H(j\omega) = H(j4) = \frac{1}{(j4+1)(j4+2)} = \frac{1}{-14 + j12}$$

故强制分量相量

$$\dot{U}_{op} = H(j4)\dot{U}_i = 2 \text{ V}$$

强制分量为

$$u_{op}(t) = 2\sqrt{2} \cos(4t) \text{ V} \quad (2)$$

由响应的初始条件及式(1)和(2)得:

$$\begin{cases} u(0_+) = 2\sqrt{2} + A + B = 0 \\ \left. \frac{du}{dt} \right|_{t=0_+} = -A - 2B = 0 \end{cases}$$

解得

$$\begin{cases} A = -4\sqrt{2} \\ B = 2\sqrt{2} \end{cases} \quad (3)$$

将式(2)、(3)代入式(1)得全响应

$$u_o = 2\sqrt{2} \cos(4t) - 4\sqrt{2}e^{-t} + 2\sqrt{2}e^{-2t} \text{ V} \quad (t > 0)$$

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10.2

$$10.2 \quad I_1 = Y_{11} U_1 + Y_{12} U_2$$

$$I_2 = Y_{21} U_1 + Y_{22} U_2 \quad Y_{12} = Y_{21}$$

$$(b) \quad -j = Y_{22}$$

$$(a) \quad -5j = Y_{21} \cdot 2j + (-j) \cdot 10j$$

$$\Rightarrow Y_{21} = Y_{12} = -2.5 + 5j$$

$$2 = Y_{11} \cdot 2j + (-2.5 + 5j) \cdot 10j$$

$$\Rightarrow Y_{11} = 12.5 - 26j$$

$$Y = \begin{bmatrix} 12.5 - 26j & -2.5 + 5j \\ -2.5 + 5j & -j \end{bmatrix}$$

日期:

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10.3(a)

解 (a): 按网孔列写 KVL 方程得

$$\begin{cases} (R+2R)I_1 + 2RI_2 = U_1 & (1) \end{cases}$$

$$\begin{cases} 2RI_1 + (R+2R)I_2 = U_2 + 3U_1 & (2) \end{cases}$$

将式(1)代入式(2)整理得

$$\begin{cases} U_1 = 3RI_1 + 2RI_2 \\ U_2 = -7RI_1 - 3RI_2 \end{cases}$$

所以

$$\mathbf{Z} = \begin{bmatrix} 3R & 2R \\ -7R & -3R \end{bmatrix}$$

日期:

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/0.5

解: 当开关断开时

$$U_1 = 5\text{V}, \quad I_1 = \frac{U_3 - U_1}{4\Omega} = 1\text{A}, \quad I_2 = 0, \quad U_2 = 3\text{V}$$

由传输参数方程得:

$$\begin{cases} 5 = A_{11} \times 3 \\ 1 = A_{21} \times 3 \end{cases} \Rightarrow \begin{cases} A_{11} = 5/3 \\ A_{21} = 1/3 \end{cases}$$

当开关接通时

$$U_1 = 4\text{V}, \quad I_1 = \frac{U_3 - U_1}{4\Omega} = 1\text{A}, \quad U_2 = 2\text{V}, \quad -I_2 = \frac{U_2}{6\Omega} = \frac{1}{3}\text{A}$$

由参数方程又得

$$\begin{cases} 4 = \frac{5}{3} \times 2 + \frac{1}{3} \times A_{12} \\ 1 = \frac{1}{3} \times 2 + \frac{1}{3} \times A_{22} \end{cases} \Rightarrow \begin{cases} A_{12} = 2 \\ A_{22} = 1 \end{cases}$$

所以

$$A = \begin{bmatrix} 5/3 & 2\Omega \\ 1/3\text{S} & 1 \end{bmatrix}$$

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10.6.6b)

根据 KVL 和理想变压器方程得

$$\begin{cases} u_1 = R_1 i_1 + n u'_2 & (1) \end{cases}$$

$$\begin{cases} i_2 = -n i_1 & (2) \end{cases}$$

$$\begin{cases} u'_2 = u_2 - R_2 i_2 & (3) \end{cases}$$

式(3)及式(2)代入式(1)整理得

$$\begin{cases} u_1 = (R_1 + n^2 R_2) i_1 + n u_2 \\ i_2 = -n i_1 \end{cases}$$

所以

$$\mathbf{H} = \begin{bmatrix} R_1 + n^2 R_2 & n \\ -n & 0 \end{bmatrix}$$

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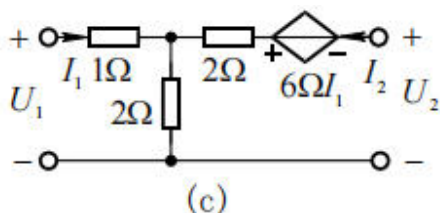
10.9(c)

(c)所示矩阵不是对称矩阵，对应的二端口方程可写成如下形式

$$\left. \begin{aligned} U_1 &= 3I_1 + 2I_2 \\ U_2 &= 2I_1 + 4I_2 - 6I_2 \end{aligned} \right\}$$

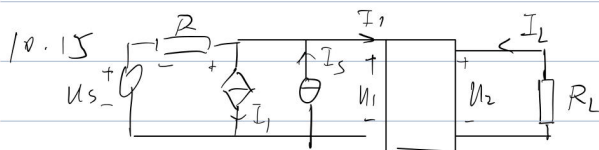
虚线左侧部分可用T形电路等效， $6I_1$ 用一个电流控制电压源表示，如图(c)

示。



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10.15

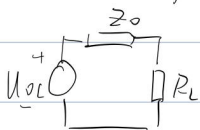


$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{cases} u_1 = u_2 - 2I_2 \\ I_1 = 3u_2 - 4I_2 \end{cases}$$

$$2I_1 = I_s - \frac{u_1 - u_s}{R} - I_1$$

$$\therefore u_2 = \frac{2}{7} + \frac{10}{7}I_2 = u_0 + z_0 I_2$$



$$R_L = z_0 = \frac{10}{7} \Omega$$

$$P_{\max} = \frac{u_0^2}{4z_0} = \frac{1}{70} \text{ W}$$

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10.18

解: 由混和参数方程得:

$$U_1 = 16I_1 + 3U_2 \quad (1)$$

$$I_2 = -2I_1 + 0.01U_2 \quad (2)$$

输入和输出端口还需满足

$$U_1 = U_s - 4I_1 \quad (3)$$

$$I_2 = -0.04U_2 \quad (4)$$

式 (1) ~ (4) 联立解得

$$U_1 = \frac{34}{35}U_s, \quad I_1 = \frac{1}{140}U_s$$

$$U_2 = \frac{2}{7}U_s, \quad I_2 = -\frac{2}{175}U_s$$

所以

$$\frac{U_2}{U_1} = \frac{5}{17}, \quad \frac{I_2}{I_1} = -1.6$$

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| 2.2 |

解: (1) 根据 Z 参数可列如下方程

$$\begin{cases} U_1(s) = \frac{2}{s+1} I_1(s) + \frac{1}{s+1} I_2(s) = U_s(s) - 1\Omega \times I_1(s) = \frac{1}{s} - I_1(s) \\ U_2(s) = \frac{1}{s+1} I_1(s) + \frac{3}{s+1} I_2(s) = -2\Omega \times I_2(s) \end{cases}$$

解得

$$I_2(s) = -\frac{0.5(s+1)}{s(s^2+5.5s+7)}$$

$$U_2(s) = -2I_2(s) = \frac{s+1}{s(s^2+5.5s+7)} = \frac{1/7}{s} + \frac{1/3}{s+2} + \frac{-10/21}{s+3.5}$$

$$u_2(t) = \left(\frac{1}{7} + \frac{1}{3}e^{-2t} - \frac{10}{21}e^{-3.5t}\right)\varepsilon(t) \text{ V}$$

(2) 由 $U_2(s)$ 的表达式可得网络函数

$$H(s) = \frac{U_2(s)}{U_s(s)} = \frac{s+1}{s^2+5.5s+7}, \quad H(j3) = H(s)|_{s=j3} = \frac{1+j3}{-2+j16.5}$$

当 $\dot{U}_{sm} = 10\angle 60^\circ \text{ V}$ 时

$$\dot{U}_{2m} = H(j3)\dot{U}_{sm} = \frac{1+j3}{-2+j16.5} \times 10\angle 60^\circ \text{ V} = 1.903\angle 34.65^\circ \text{ V}$$

所以

$$u_2(t) = 1.903\cos(3t + 34.65^\circ) \text{ V}$$

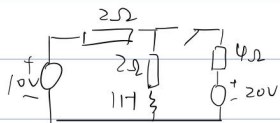
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10.23

$$10.23 \quad \begin{cases} I_1 = U_1 - \frac{1}{2}U_2 \\ I_2 = -\frac{1}{2}U_1 + \frac{2}{3}U_2 \end{cases}$$

$$I_1 = \frac{30 - U_1}{2} \Rightarrow U_2 = 10 + 5I_2$$



$$U_1' \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{4} \right) = \frac{10}{2} + \frac{20}{4}$$

$$\Rightarrow U_1' = 8V, \quad i(0^-) = 4A$$

$$i(+\infty) = 2.5A, \quad \tau = \frac{L}{R} = \frac{1}{4}$$

$$\therefore i = (2.5 + 1.5e^{-4t})A$$

$$U_2 = L \frac{di}{dt} + 2i = (5 - 3e^{-4t})V$$